

The Three-Twentieths Common-Anchor Conjecture

A single, self-contained open question in quartic zero-repair transport

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Abstract

We isolate one precise open question from the quartic zero-repair problem. The question is self-contained and requires only the definitions given here.

For the explicit laws

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right),$$

we first determine the sharp threshold inside the one-step Bernoulli reserve family

$$R_\varepsilon = (1 - \varepsilon)\delta_0 + \varepsilon\delta_1.$$

An exact rational calculation gives

$$Q * R_\varepsilon \succeq_{\text{st}} P \iff \varepsilon \geq \frac{3}{20}.$$

Thus

$$R_* := \frac{17}{20}\delta_0 + \frac{3}{20}\delta_1$$

is the least-mean reserve in this Bernoulli family that universally insures Q against P .

The open question asks whether, for every depth n , one can choose an arbitrarily near-optimal exact depth- n zero-repair tree whose last generation admits a *single common viable anchor* after convolution by R_* . If this conjecture holds at every scale, then

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + \frac{3}{20}$$

for every n , and therefore

$$\mathcal{B}_n = O(\log n).$$

The exact-tree common-anchor conjecture stated here has not yet been tested numerically in either direction. Earlier finite linear-program experiments on positive last-layer normal forms concern a related block-relaxed problem and are included only as motivation, not as evidence for the conjecture itself.

1 The question in one page

The problem has a simple operational interpretation.

A law B is called depth- n viable if it can serve as the root of an n -generation positive zero-repair tree. At every node the recursion is

$$Q * B_h = \sum_{x=0}^4 P(x) \delta_x * B_{hx}.$$

Thus the law $Q * B_h$ must be split into five positive child laws with prescribed P -weights and prescribed shifts.

Suppose an exact depth- n tree has already been built. At its final internal layer, $|h| = n - 1$, we would like to replace the boundary children by new children that are themselves viable for another n generations. Doing this simultaneously for every last-layer history would concatenate two depth- n constructions into one depth- $2n$ construction.

The reserve

$$R_* = \frac{17}{20}\delta_0 + \frac{3}{20}\delta_1$$

is special because

$$Q * R_* \succeq_{\text{st}} P.$$

If the same reserve is convolved through the whole already constructed prefix, all old zero-repair equalities remain exact. Hence only the final internal layer must be rebuilt.

The conjecture asks whether one can choose the original near-optimal depth- n tree so that there is one law

$$C_n \in \mathcal{K}_n$$

with

$$(Q * B_h) * R_* \succeq_{\text{st}} P * C_n \quad \text{for every } |h| = n - 1.$$

If so, monotone coupling produces new depth- n -viable children at every last-layer history. The entire doubling costs only

$$\mathbb{E}R_* = \frac{3}{20}.$$

Thus the open question is not whether one can repair each history separately. It is whether *one common viable anchor* can sit below all insured last-layer parents simultaneously.

2 Formal setup

Fix

$$\mathcal{A} = \{0, 1, 2, 3, 4\},$$

and define probability laws

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64}\right). \quad (1)$$

Their probability-generating functions satisfy

$$\boxed{G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.} \quad (2)$$

For probability laws μ, ν on $\mathbb{Z}_{\geq 0}$, write

$$\mu \succeq_{\text{st}} \nu$$

for the usual stochastic order:

$$\mu \succeq_{\text{st}} \nu \iff F_\mu(t) \leq F_\nu(t) \quad \text{for every } t \in \mathbb{Z}. \quad (3)$$

Equivalently, there is a coupling

$$(X, Y), \quad X \sim \mu, \quad Y \sim \nu,$$

such that $X \geq Y$ almost surely; see Strassen's theorem [1] and the standard theory of stochastic orders [2]. Convolution preserves the order:

$$\mu \succeq_{\text{st}} \nu \implies \mu * \lambda \succeq_{\text{st}} \nu * \lambda.$$

Definition 1 (Depth- n zero-repair tree). *A depth- n zero-repair tree is a family of probability laws*

$$\{B_h\}_{|h| \leq n}$$

on $\mathbb{Z}_{\geq 0}$, indexed by words

$$h = (x_1, \dots, x_t) \in \mathcal{A}^t,$$

such that

$$\boxed{Q * B_h = \sum_{x \in \mathcal{A}} P(x) \delta_x * B_{hx}} \quad (|h| < n). \quad (\text{ZR})$$

Definition 2 (Viability sets). *Let*

$$\mathcal{K}_n$$

be the set of all probability laws B on $\mathbb{Z}_{\geq 0}$ that occur as the root $B_{\emptyset}$ of some depth- n zero-repair tree. Put

$$\mathcal{K}_0 := \{\text{all probability laws on } \mathbb{Z}_{\geq 0}\}.$$

The finite-horizon bill is

$$\boxed{\mathcal{B}_n := \inf_{B \in \mathcal{K}_n} \mathbb{E}B,} \quad (4)$$

where

$$\mathbb{E}B := \sum_{j \geq 0} j B(j).$$

Since

$$\mathcal{K}_{n+1} \subseteq \mathcal{K}_n,$$

the sequence $(\mathcal{B}_n)$ is nondecreasing.

3 The sharp Bernoulli insurance threshold

For $0 \leq \varepsilon \leq 1$, define the Bernoulli reserve

$$R_\varepsilon := (1 - \varepsilon)\delta_0 + \varepsilon\delta_1, \quad G_{R_\varepsilon}(z) = 1 - \varepsilon + \varepsilon z. \quad (5)$$

Theorem 3 (Sharp Bernoulli insurance threshold). *For the pair (P, Q) in (1),*

$$\boxed{Q * R_\varepsilon \succeq_{\text{st}} P \iff \varepsilon \geq \frac{3}{20}.} \quad (6)$$

Consequently the least-mean reserve in the Bernoulli family (5) is

$$\boxed{R_* = \frac{17}{20}\delta_0 + \frac{3}{20}\delta_1, \quad \mathbb{E}R_* = \frac{3}{20}.} \quad (7)$$

At the critical value,

$$\boxed{\frac{G_P(z) - G_Q(z)G_{R_*}(z)}{1 - z} = \frac{41 + 90z^2 + 40z^3 + 21z^4}{1280}.} \quad (8)$$

Proof. A direct calculation gives

$$\begin{aligned} \frac{G_P(z) - G_Q(z)G_{R_\varepsilon}(z)}{1 - z} &= \frac{1}{64} \left[(1 + 7\varepsilon) + (-3 + 20\varepsilon)z \right. \\ &\quad \left. + (3 + 10\varepsilon)z^2 + (-1 + 20\varepsilon)z^3 + 7\varepsilon z^4 \right]. \end{aligned} \quad (9)$$

For any law μ on $\mathbb{Z}_{\geq 0}$,

$$\sum_{t \geq 0} F_\mu(t) z^t = \frac{G_\mu(z)}{1 - z}.$$

Hence

$$Q * R_\varepsilon \succeq_{\text{st}} P$$

is equivalent to coefficientwise nonnegativity of

$$\frac{G_P - G_Q G_{R_\varepsilon}}{1 - z}.$$

For $0 \leq \varepsilon \leq 1$, all coefficients in (9) are nonnegative if and only if

$$-3 + 20\varepsilon \geq 0.$$

Thus

$$\varepsilon \geq \frac{3}{20}.$$

Substitution of $\varepsilon = 3/20$ yields (8). □

Remark 4 (Meaning of sharpness). Theorem 3 is sharp *within the one-step Bernoulli family*

$$(1 - \varepsilon)\delta_0 + \varepsilon\delta_1.$$

No claim is made here that $3/20$ is the smallest possible mean among *all* probability reserves R satisfying

$$Q * R \succeq_{\text{st}} P.$$

Corollary 5 (Universal insured dominance). *For every probability law B on $\mathbb{Z}_{\geq 0}$,*

$$\boxed{Q * (B * R_*) \succeq_{\text{st}} P * B.} \tag{10}$$

Proof. By associativity and commutativity of convolution,

$$Q * (B * R_*) = B * (Q * R_*).$$

Theorem 3 gives

$$Q * R_* \succeq_{\text{st}} P,$$

and monotonicity of convolution gives

$$B * (Q * R_*) \succeq_{\text{st}} B * P.$$

□

4 A structural fact about the viability sets

The common-anchor argument uses the following order property.

Proposition 6 (Stochastic upward closure). *For every $n \geq 0$,*

$$C \in \mathcal{K}_n, \quad D \succeq_{\text{st}} C \quad \implies \quad D \in \mathcal{K}_n.$$

Proof. We argue by induction on n .

For $n = 0$ there is nothing to prove. Assume the result holds for $\mathcal{K}_n$, and let

$$C \in \mathcal{K}_{n+1}, \quad D \succeq_{\text{st}} C.$$

Choose children $C_x \in \mathcal{K}_n$ satisfying

$$Q * C = \sum_x P(x) \delta_x * C_x.$$

Realize the right-hand side by first sampling

$$X \sim P$$

and then, conditional on $X = x$, sampling

$$J_x \sim C_x.$$

Thus

$$S_C := X + J_X \sim Q * C.$$

Since

$$D \succeq_{\text{st}} C,$$

convolution gives

$$Q * D \succeq_{\text{st}} Q * C.$$

By Strassen's theorem [1], there is a monotone coupling

$$S_D \sim Q * D, \quad S_C \sim Q * C, \quad S_D \geq S_C \quad \text{a.s.}$$

Using a monotone coupling kernel conditional on S_C , we may combine this with the preceding representation of $S_C = X + J_X$.

For each x , define

$$D_x := \text{Law}(S_D - X \mid X = x).$$

Then

$$S_D - X \geq S_C - X = J_x,$$

so

$$D_x \succeq_{\text{st}} C_x.$$

By the induction hypothesis,

$$D_x \in \mathcal{K}_n.$$

Finally,

$$Q * D = \text{Law}(S_D) = \sum_x P(x) \delta_x * D_x,$$

so $D \in \mathcal{K}_{n+1}$. □

5 The common-anchor conjecture

We now state the open question in a form that does not require existence of an exact optimizer for $\mathcal{B}_n$.

Conjecture 7 (Three-twentieths common anchor). *For every $n \geq 1$ and every $\eta > 0$, there exist*

(i) *a depth- n zero-repair tree*

$$\{B_h\}_{|h| \leq n}$$

with

$$\mathbb{E}B_\emptyset \leq \mathcal{B}_n + \eta, \quad (11)$$

and

(ii) *a single law*

$$C_n \in \mathcal{K}_n$$

such that

$$\boxed{(Q * B_h) * R_* \succeq_{\text{st}} P * C_n} \quad \text{for every } h \text{ with } |h| = n - 1, \quad (12)$$

where

$$R_* = \frac{17}{20}\delta_0 + \frac{3}{20}\delta_1.$$

The use of an arbitrarily near-optimal tree rather than an exact optimizer is deliberate: it avoids any unnecessary attainment assumption in the countable-state optimization problem.

Remark 8 (A simpler sufficient condition). A stronger, simpler condition would be the existence of

$$C_n \in \mathcal{K}_n$$

such that

$$B_h \succeq_{\text{st}} C_n \quad \text{for every } |h| = n - 1.$$

Indeed, Corollary 5 would give

$$(Q * B_h) * R_* \succeq_{\text{st}} P * B_h \succeq_{\text{st}} P * C_n.$$

Thus a common stochastic floor would imply Conjecture 7.

6 Why the conjecture would close the dyadic upper bound

Theorem 9 (Conditional dyadic closure). *If Conjecture 7 holds for every n , then*

$$\boxed{\mathcal{B}_{2n} \leq \mathcal{B}_n + \frac{3}{20}} \quad (n \geq 1). \quad (13)$$

Consequently

$$\boxed{\mathcal{B}_n = O(\log n)}. \quad (14)$$

Proof. Fix n and $\eta > 0$. Choose the depth- n tree

$$\{B_h\}_{|h| \leq n}$$

and common anchor $C_n \in \mathcal{K}_n$ supplied by Conjecture 7.

For every prefix law through depth $n - 1$, set

$$\widehat{B}_h := B_h * R_*, \quad |h| \leq n - 1.$$

If $|h| \leq n - 2$, then the original tree satisfies

$$Q * B_h = \sum_x P(x) \delta_x * B_{hx}.$$

Convolving by the same reserve R_* gives

$$Q * \widehat{B}_h = \sum_x P(x) \delta_x * \widehat{B}_{hx}.$$

Thus every zero-repair equality above the last internal layer remains exact.

Now fix a history h with $|h| = n - 1$. By the anchor condition,

$$(Q * B_h) * R_* \succeq_{\text{st}} P * C_n.$$

Let

$$S \sim (Q * B_h) * R_*.$$

Independently sample

$$X \sim P, \quad J \sim C_n,$$

so that

$$X + J \sim P * C_n.$$

By Strassen's theorem there is a coupling such that

$$S \geq X + J \quad \text{a.s.}$$

For each $x \in \mathcal{A}$, define

$$D_{hx} := \text{Law}(S - X \mid X = x).$$

Then

$$D_{hx} \succeq_{\text{st}} C_n.$$

Since

$$C_n \in \mathcal{K}_n,$$

Proposition 6 implies

$$D_{hx} \in \mathcal{K}_n.$$

Moreover

$$(Q * B_h) * R_* = \sum_x P(x) \delta_x * D_{hx}.$$

Hence the insured prefix can be completed at depth n by children D_{hx} , each of which admits a further exact depth- n continuation.

Appending one such continuation below every depth- n child produces an exact depth- $2n$ zero-repair tree. Its root law is

$$B_{\emptyset} * R_*,$$

so

$$\mathcal{B}_{2n} \leq \mathbb{E}B_\emptyset + \mathbb{E}R_* \leq \mathcal{B}_n + \eta + \frac{3}{20}.$$

Since $\eta > 0$ was arbitrary,

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + \frac{3}{20}.$$

For dyadic $n = 2^k$,

$$\mathcal{B}_{2^k} \leq \mathcal{B}_1 + \frac{3k}{20}.$$

For arbitrary n , choose

$$k = \lceil \log_2 n \rceil.$$

Monotonicity of $\mathcal{B}_n$ gives

$$\mathcal{B}_n \leq \mathcal{B}_{2^k} \leq \mathcal{B}_1 + \frac{3}{20} \lceil \log_2 n \rceil,$$

which proves (14). □

Remark 10 (A bounded near-optimality loss). If one can prove only a version of the conjecture with

$$\mathbb{E}B_\emptyset \leq \mathcal{B}_n + C_0$$

for a fixed constant C_0 , the same construction gives

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C_0 + \frac{3}{20},$$

which still implies

$$\mathcal{B}_n = O(\log n).$$

7 An explicit viability barrier

The sets $\mathcal{K}_n$ are not featureless. The following exact inequality is a simple example of a recursively inherited viability constraint.

Define

$$a_0 := 1, \quad a_{k+1} := \frac{8 + 16a_k}{27}. \tag{15}$$

Proposition 11 (Two-coordinate viability barrier). *Every*

$$B \in \mathcal{K}_k$$

satisfies

$$\boxed{B(0) + \frac{7}{27}B(1) \leq a_k.} \tag{16}$$

Moreover

$$\boxed{a_k = \frac{8}{11} + \frac{3}{11} \left(\frac{16}{27} \right)^k,} \tag{17}$$

and therefore

$$a_k \downarrow \frac{8}{11}.$$

Proof. For $k = 0$, normalization gives

$$B(0) + \frac{7}{27}B(1) \leq 1 = a_0.$$

For the first nontrivial step, if $B \in \mathcal{K}_1$, then one-step viability implies

$$Q * B \succeq_{\text{st}} P.$$

At threshold $t = 1$,

$$F_{Q*B}(1) = \frac{27B(0) + 7B(1)}{64} \leq F_P(1) = \frac{3}{8}.$$

Thus

$$B(0) + \frac{7}{27}B(1) \leq \frac{8}{9} = a_1.$$

Now assume (16) holds on $\mathcal{K}_k$, and let

$$B \in \mathcal{K}_{k+1}.$$

Choose children

$$C_x \in \mathcal{K}_k$$

satisfying the zero-repair identity. Comparing the coefficient of 0 gives

$$\frac{7}{64}B(0) = \frac{1}{8}C_0(0),$$

hence

$$C_0(0) = \frac{7}{8}B(0).$$

Comparing the coefficient of 1 gives

$$\frac{20B(0) + 7B(1)}{64} = \frac{1}{8}C_0(1) + \frac{1}{4}C_1(0).$$

Since

$$C_0(1) \leq 1 - C_0(0) = 1 - \frac{7}{8}B(0)$$

and, by the induction hypothesis,

$$C_1(0) \leq a_k,$$

we obtain

$$20B(0) + 7B(1) \leq 8 - 7B(0) + 16a_k.$$

Therefore

$$27B(0) + 7B(1) \leq 8 + 16a_k,$$

or

$$B(0) + \frac{7}{27}B(1) \leq \frac{8 + 16a_k}{27} = a_{k+1}.$$

The affine recursion (15) has fixed point $8/11$, and solving it gives (17). $\square$

Remark 12. Proposition 11 supplies only one family of necessary inequalities. It is not claimed to characterize $\mathcal{K}_n$, nor is it claimed to be a complete facet description.

8 Relation to the earlier last-layer experiments

The exact-tree conjecture in Section 5 should be distinguished from the finite block-relaxed calculations that motivated it.

In the preceding research note [6], a block relaxation was used in which internal prefix equalities were temporarily removed. A separate linear program then asked whether those residuals could be pushed to the final independent prefix layer while retaining positive laws. Representative finite-support results were

(n, M)	unconstrained positive filling	positive last-layer filling
(3, 8)	0.0697761219022406	0.0697761219025007
(3, 10)	0.0697761218851941	0.0697761218923195
(4, 8)	0.0739180923076735	0.0739182643909350.

These computations support a *positive last-layer normal-form* phenomenon in those finite truncations. They do not test Conjecture 7. The present conjecture begins instead with an already exact depth- n tree and asks for a common viable anchor after insurance by R_* .

Accordingly:

the exact-tree common-anchor conjecture is currently numerically untested.

The earlier calculations are motivation for the architecture, not evidence for the truth of the new conjecture.

9 How the conjecture can be attacked

There are two logically independent routes.

Construction

For a given n , find a depth- n tree close to $\mathcal{B}_n$ and a law

$$C_n \in \mathcal{K}_n$$

such that

$$(Q * B_h) * R_* \succeq_{\text{st}} P * C_n$$

for every $|h| = n - 1$.

A particularly attractive sufficient mechanism is a common stochastic floor:

$$B_h \succeq_{\text{st}} C_n \quad \forall |h| = n - 1.$$

At a fixed support cutoff M , this becomes a finite system of linear equalities and CDF inequalities and can therefore be attacked by linear programming. Passing from finite cutoffs to the unrestricted countable-state statement requires a separate support, tightness, or compactness argument.

Counterexample

Find an n for which every near-optimal depth- n tree fails the common anchor condition. One possible certificate would exhibit incompatible last-layer histories: insured laws whose common stochastic lower bounds lie outside $\mathcal{K}_n$.

At a finite support cutoff such a failure can be certified by LP duality. An unrestricted counterexample would additionally have to exclude the possibility that mass escaping beyond the cutoff restores feasibility.

10 Precise status

Statement	Status
Quartic identity $G_P - G_Q = (1 - z)^4/64$	Proved
Sharp Bernoulli threshold $Q * R_\varepsilon \succeq_{\text{st}} P \iff \varepsilon \geq 3/20$	Proved exactly
Universal insured dominance $Q * (B * R_*) \succeq_{\text{st}} P * B$	Proved
Stochastic upward closure of $\mathcal{K}_n$	Proved
Two-coordinate viability barrier	Proved
Positive last-layer normal form in earlier finite block-relaxed LPs	Numerical motivation
Exact-tree common-anchor condition at any n	Open; not yet tested
Conjecture 7 uniformly for all n	Open
Dyadic inequality $\mathcal{B}_{2n} \leq \mathcal{B}_n + 3/20$	Conditional on the conjecture
Global $O(\log n)$ upper bound	Conditional on the conjecture

11 Conclusion

The open question can be stated in one line:

$$\begin{aligned} &\forall n, \forall \eta > 0, \exists \text{ a depth-}n \text{ tree with } \mathbb{E}B_\emptyset \leq \mathcal{B}_n + \eta, \\ &\exists C_n \in \mathcal{K}_n \text{ such that } (Q * B_h) * R_* \succeq_{\text{st}} P * C_n \quad \forall |h| = n - 1, \\ &R_* = \frac{17}{20}\delta_0 + \frac{3}{20}\delta_1. \end{aligned}$$

The reserve is not fitted from numerics: its coefficient $3/20$ is the exact sharp threshold in the Bernoulli family. The only unresolved point is the simultaneous common-anchor geometry of near-optimal exact trees.

A proof would give

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + \frac{3}{20}$$

and hence a logarithmic upper bound. A counterexample at any scale would falsify this particular mechanism and identify where the common-anchor picture breaks.

Either outcome is mathematically informative.

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